EMI Immunity of the Nauta Inverter-Based Amplifier

A. Rosa, *Member, IEEE*, A. Richelli, *Senior Member, IEEE*, L. Colalongo, *Member, IEEE*,

***Abstract*— This paper investigates the effect of Electromagnetic Interferences (EMI) on inverter-based analog amplifiers, as the Nauta operational transconductance amplifier (OTA). We show that, thanks to the their simple circuital topology that does not include current mirrors and differential pair, the inverter-based analog amplifiers have a much higher EMI immunity: about one order of magnitude.**

Index Terms—**Electromagnetic immunity, analog integrated circuits, inverter-based amplifiers, Nauta OTA, CMOS amplifiers.**

I. Introduction

Inverter-based amplifiers, as the Nauta OTA [1], take advantage of recent integrated technologies compared to the conventional analog topologies that, on the other hand, become more difficult to design. Several topologies have stemmed from the Nauta OTA: they are used as active filters [2]- [3], fully differential amplifiers [4]- [5], hybrid architectures [6]- [7], etc., showing the interest of the research community to this kind of circuits. In this correspondence paper, in addition to the conventional figure of merit (gain, gainbandwidth product, power consumption, common mode range), for the first time in the authors' knowledge, the susceptibility to the EMI of the Nauta-derived OTA is investigated and compared with the well known analog Miller topology. The vulnerability of electronic circuits to the EMI is indeed a critical issue [8], since nowadays electronic is ubiquitous and the number of possible sources, victims, and paths of EMI is continuously increasing, [9]. Moreover, among the electronic circuits, the analog ones and, in particular, the operational amplifiers, are the most sensitive to interferences. EMI are detrimental not only on the target amplifier but on overall system as well [10] - [12].

II. Nauta-derived single-end operational transconductance amplifier

The Nauta OTA was conceptualized in [1] and originally designed as a fully-differential amplifier: conceptually it is a simple transconductor based on CMOS inverters. The CMOS inverter, in fact, has no internal nodes, and, if the transistors operate in strong inversion and β is perfectly matched, has excellent high frequency behavior and good linearity in V-I conversion. Furthermore, several new topologies were proposed in the recent literature to overcome process, voltage, and temperature (PVT) variations, to improve the linearity,

A. Rosa, A. Richelli and L. Colalongo are with the Department of Information Engineering, University of Brescia, Italy, Email: anna.richelli@unibs.it

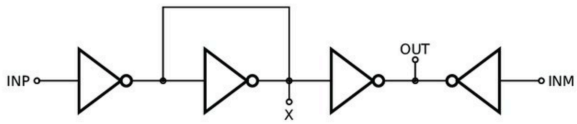

Fig. 1. Schematic of the single-ended inverter-based amplifier [13].

the gain, and to achieve a better common mode control [2]-[5], [13]. Among them, it is worth remembering, the single-ended topologies [13], [14]. For the sake of simplicity, in this work, we have designed and investigated the single-ended inverter-based amplifier (SEIA) of (Fig.1) [13], where all the transconductance stages are CMOS inverters, as in the Nauta OTA [1]. The SEIA is designed in UMC 180nm technology. To straightforwardly compare the first stage of the Miller amplifier with the SEIA the transistors are dimensioned to have a gain of about 40dB. The transistors sizing and capacitive load are listed in Tab. I. The AC characteristics in Tab. II. In addition, several simulations were also performed at much lower power supply to investigate the suitability of SEIA for ultra-low power applications. Corner simulations were also worked out: they do not show any critical issue. The gain of the SEIA ranges from 38dB to 42dB, depending on corner and voltage supply (0.5V, 1V and 1.8V). The worst case (34dB) is reached at Vdd=0.5V and fast NMOS slow PMOS (FNSP). The phase margin remains of about 70° in all simulations. The GBW ranges between 362kHz-1.52Mhz at Vdd=0.5V, 24MHz-42.5MHz at Vdd=1V, and 128MHz-180MHz at Vdd=1.8V. It is worth adding that the SEIA is dimensioned at Vdd=1V and not deeply optimized: the same sizing is used at Vdd=0.5V and 1.8V.

TABLE I
Devices' Sizing of the Inverter-Based OTA

NMOS	W=48μm	L=4μm
PMOS	W=280μm	L=4μ
Cload	1pF	

RISULTATI AC NAUTA

TABLE II

AC SIMULATIONS OF THE INVERTER-BASED OTA

Vdd	A_v	GBW	PM	Power	CM range
1.8V	38dB	155MHz	80°	2.9mW	550mV
1V	40dB	32MHz	75°	118μW	400mV
0.5V	38.5dB	770kHz	73°	6μW	300mV

III. Comparison with Miller OTA

To get insight on the EMI susceptibility of the single-ended inverter-based OTA, it is compared with the well-known and deeply investigated (also in terms of EMC) Miller amplifier. In particular, the first stage of the Miller amplifier, based on a differential pair and a current mirror active load, is compared with the SEIA. To fairly compare the EMI susceptibility of both amplifiers, the first stage of the Miller amplifier is designed to have AC characteristics similar to that of the inverter-based: the characteristics are listed in Tab. III. As

RISULTATI AC MILLER

TABLE III

AC SIMULATIONS OF THE FIRST STAGE OF MILLER AMPLIFIER

Vdd	A_v	GBW	PM	Power	CM range
1.8V	38dB	148MHz	75°	2.4mW	1.4V

customary in the literature [9], the interfering signals are applied to the non inverting input and the amplifiers are connected in voltage follower configuration; the amplitude of the interferences is rail to rail (i.e. a large sinusoidal signal, from Gnd to Vdd, centered in the middle of the input common mode range) and the frequency ranges from several MHz up to Ghz to span the spectrum of all the possible disturbing signals. The simulations of the post layout of both the amplifiers are shown in Fig. 2. One can see at a glance that the SEIA exhibits a remarkably higher immunity to the interferences with respect to the Miller amplifier. This result can be easy explained by comparing the slew rate and the schematics of both amplifiers. In fact, the EMI susceptibility is related to the slew rate asymmetry, to the non linear distortion of the differential stage, and to the charge pumping in the current mirrors. The inverter based amplifier, thanks to its simple topology and the absence of internal nodes, has symmetrical positive and negative slew rate. It is the key to understand the EMI immunity with respect to the Miller amplifier that, due to the non ideality of the differential pair [16] combined with the EMI susceptibility of the current mirror [17], typically exhibits an asymmetric response to sharp positive and negative input voltage variation.

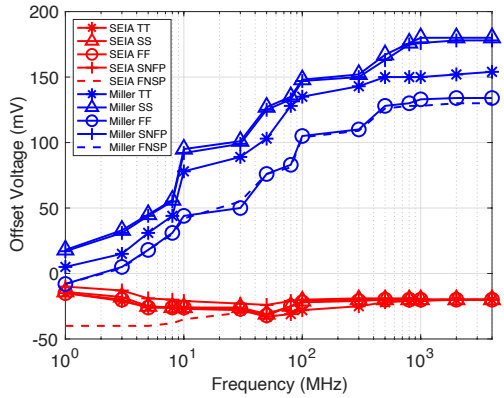

Fig. 2. EMI induced offset: comparison between inverter-based and Miller amplifier.

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